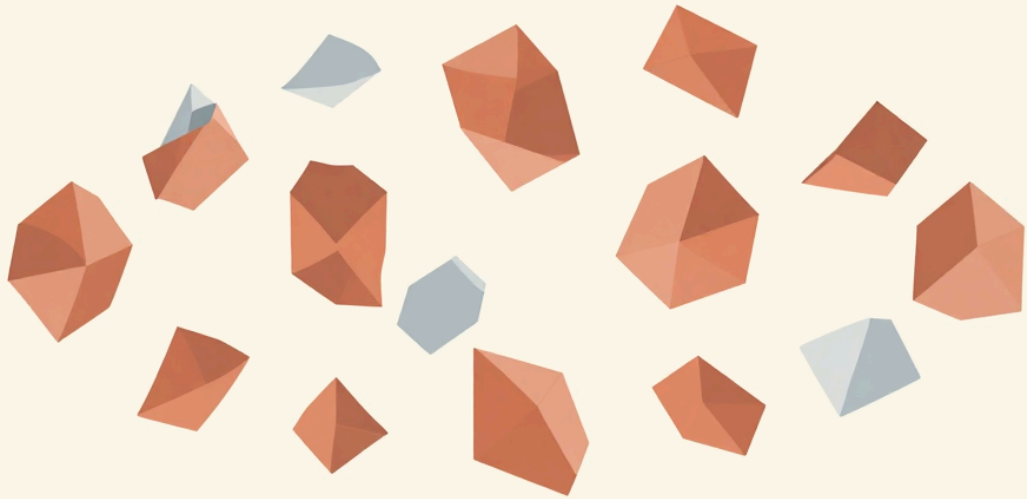




Attention is All You Need

PAPER EXPLAINED

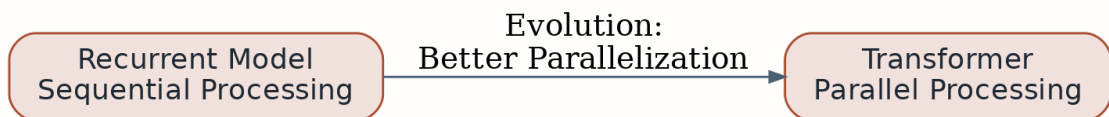
Attention Is All You Need: The End of Recurrence

Attention Is All You Need

<https://arxiv.org/abs/1706.03762>

The Parallelization Bottleneck

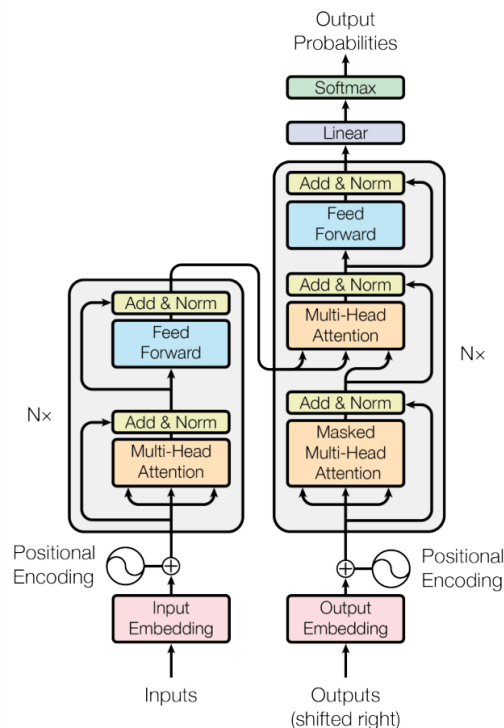
- Recurrent models preclude parallelization within training examples due to their sequential nature.
- Transformers allow significantly more parallelization, reaching state of the art in twelve hours on eight P100 GPUs.



Transformers enable parallel processing, overcoming the sequential bottleneck of recurrent models.

Transformer Architecture Details

- The Transformer eschews recurrence entirely, relying solely on attention mechanisms.
- It is the first transduction model relying entirely on self-attention.

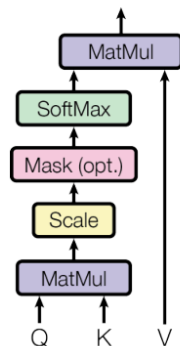


The Transformer - model architecture.

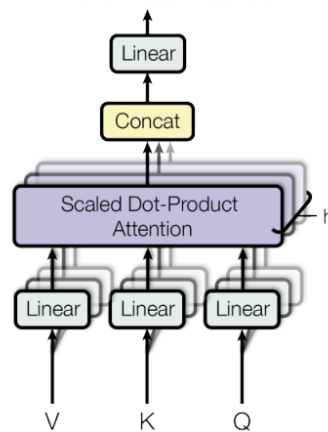
Transformer Encoder Structure

- The Transformer uses stacked self-attention and point-wise fully connected layers for both encoder and decoder.
- The encoder stacks $N=6$ identical layers, each with multi-head self-attention.
- Self-attention keys, values, and queries originate from the previous layer's output.

Scaled Dot-Product Attention



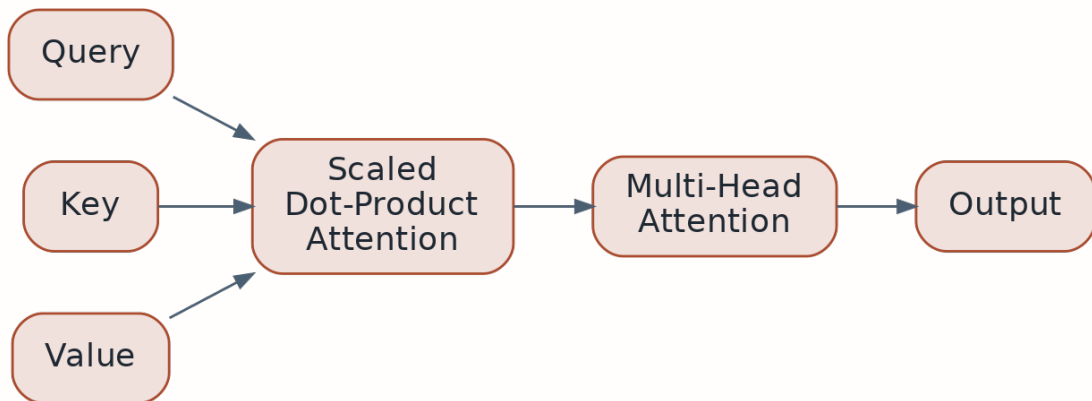
Multi-Head Attention



(left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

Multi-Head Self-Attention Mechanism

- Self-attention relates different positions of a single sequence to compute its representation.
- Multi-head attention allows the model to attend to information from different representation subspaces simultaneously.
- Decoder self-attention is masked to prevent leftward information flow and preserve the auto-regressive property.



The multi-head self-attention mechanism: queries, keys, and values are processed through scaled dot-product attention to produce the output.

Transformer Translation Results

- The Transformer reaches new state-of-the-art translation quality after training for twelve hours on eight P100 GPUs.
- The big model achieves a BLEU score of 28.4.
- The big model achieves a BLEU score of 41.0, outperforming all previously published single models.

Beyond Translation: Parsing

- Transformer achieves 92.7 F1 in semi-supervised English constituency parsing.
- It outperforms all models except Recurrent Neural Network Grammar.
- Transformer beats Berkeley-Parser using only 40K training sentences.

Limitations and Future Work

- Reducing key size d_k hurts quality, suggesting compatibility is difficult.
- Future work extends the Transformer to other modalities using local attention mechanisms.

The Transformer Takeaway

- Achieves new state-of-the-art results on WMT 2014 English-to-German and English-to-French tasks.
- Trains significantly faster than architectures based on recurrent or convolutional layers.